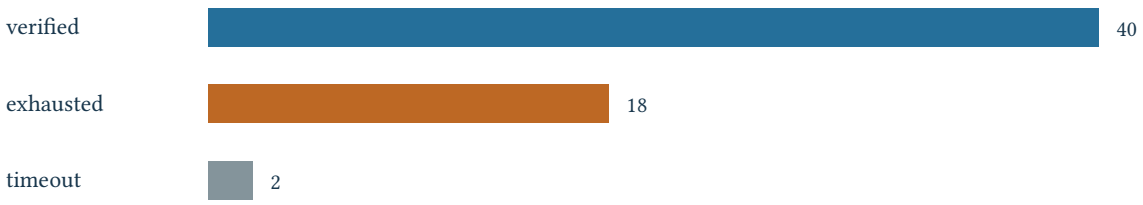
Watchdog repair: control and pilot evidence

17 October 2026. New campaign data, separate from historical dev/release runs.

Repaired runner: audited control and three-root pilot

60 attempts, two public targets per model, one round, zero warmups. Counts only; no admitted runtime comparison.



Completed pairs match baseline: control 12, pilot 46. Two pilot pairs remain censored, not equal-count comparisons.

The earlier release survey halted after 70 audited vertices on a failed standard audit. One late child message carried a checked scalar while the parent labeled the attempt timeout. Its failed 16-record capsule is retained; no historical record or auditor rule was changed.

The repaired runner makes watchdog censoring authoritative. Timed-out attempts have no accepted scalar, success check, final cost/counters or solve timing. Any parsed late message remains in the sealed diagnostic error. Solver arithmetic, public targets, factor bases and budgets are unchanged.

Fresh small control: 12 attempts. Fresh three-root pilot: 48 attempts. Every session passes the standard audit; successful and exhausted deterministic reports reproduce. Native comparison to the earlier release baseline checks method identities, exact outcomes, scalars, GAE, phases, field operations, counters and factor bases. It finds 12 matching completed control pairs and 46 matching completed pilot pairs. Two censored pilot pairs have no completed reports to compare. Elapsed values are excluded. Same-implementation replay and profile matching are not independent implementation validation.

The frozen runner hash is 35d89073f4d666b094c213a1d31b0c819b63600123e067c111af3c947d7609a9. The driver hash is b8b489c7225c64ed492ffce39286c995f5ca820960956cf706c0d3110f5bdd2d. Clean source 2bf7700b7 contains the repair. The native build source/protocol and input/binary hashes are frozen in each campaign. The separate full serial release library check passes 4,236 tests, zero failures, 117 existing ignored. The previously failing parallel Joux–Vitse case passes serially; its parallel cause remains unresolved. No solver budget or exact-answer assertion changed.

Host: Xeon Platinum 8573C, Linux x86-64, four-CPU quota, 16 GiB memory limit. Release optimization; wall observations remain unqualified. No external comparison, fastest-method claim or speedup is made. The methods retain actual largest prime orders 61751, 19768889568671 and 4253 in the pilot; the small control has order 7 over p=113. No smaller subgroup replaced the 45-bit case.

Passing these gates permits a separately frozen 24,823-vertex bounded survey, not the full eight-target/five-round/million-trial matrix. Planning is not execution. Its actual launch and current coverage require the separate launch receipt and live ledger. Original A–E completion and large-extension finite-field IC remain open. Source protocols, raw capsules and native comparison receipts are in the accompanying archive and study execution tree.

Native per-arm GAE distributions

Graph	Arm	Outcome	Attempts	Cost n	Min GAE	Mean GAE	Max GAE
control	bsgs	verified	2	2	4	4.00	4
control	projected-2-mitm	verified	2	2	120	123.50	127
control	projected-2-subtract	verified	2	2	111	115.00	119
control	raw-control	exhausted	2	2	1,327	1,335.50	1,344
control	rho	exhausted	1	1	8,013	8,013.00	8,013
control	rho	verified	1	1	23	23.00	23
control	rho-aa	exhausted	1	1	8,013	8,013.00	8,013
control	rho-aa	verified	1	1	23	23.00	23
mixed	bsgs	verified	2	2	75	104.00	133
mixed	projected-16-mitm	verified	2	2	4,242	4,419.00	4,596
mixed	projected-16-subtract	verified	2	2	10,148	12,328.00	14,508
mixed	projected-64-mitm	verified	2	2	20,463	20,467.50	20,472
mixed	projected-64-subtract	verified	2	2	22,087	22,588.00	23,089
mixed	raw-control	exhausted	2	2	13,361	13,383.00	13,405
mixed	rho	verified	2	2	138	189.50	241
mixed	rho-aa	verified	2	2	138	189.50	241
three	bsgs	timeout	2	0	missing	missing	missing
three	projected-16-mitm	exhausted	2	2	8,191	8,204.00	8,217
three	projected-16-subtract	exhausted	2	2	138,735	138,748.00	138,761
three	projected-64-mitm	exhausted	2	2	20,239	20,252.00	20,265
three	projected-64-subtract	exhausted	2	2	536,271	536,284.00	536,297
three	raw-control	exhausted	2	2	14,479	14,492.00	14,505
three	rho	verified	2	2	2,005,511	3,519,227.50	5,032,944
three	rho-aa	verified	2	2	2,005,511	3,519,227.50	5,032,944
two	bsgs	verified	2	2	266	276.50	287
two	projected-16-mitm	verified	2	2	6,493	6,927.00	7,361
two	projected-16-subtract	verified	2	2	76,008	90,295.50	104,583
two	projected-64-mitm	verified	2	2	19,530	19,970.00	20,410
two	projected-64-subtract	verified	2	2	130,718	160,668.50	190,619
two	raw-control	exhausted	2	2	13,726	13,750.00	13,774
two	rho	verified	2	2	355	360.50	366
two	rho-aa	verified	2	2	355	360.50	366

GAE with unpriced work is a lower bound. Exhaustion is an attempted-budget cost, not a successful DLP cost. Timeout counters/costs are missing, not zero. Each outcome stays separate, with present-value n. Raw phase/native counters, memory and unqualified elapsed observations remain in the native summary and CSV. Repeated baseline and candidate targets are not pooled.

Censored observations (one row per attempt)

Graph	Arm	Record id	Final GAE	Raw process s	Peak RSS KiB
three	bsgs	ECR1hdd5aa78ac1bc	missing	3.011223532	166736.0
three	bsgs	ECR1h1b9c3c99a3b7	missing	3.016101743	166728.0

These process observations are not completed solve times. The final-cost table above contains present reports only; censored costs stay missing.